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DSA210-Final Project Report

A Multivariate Analysis of My Music Consumption Patterns Based on My Menstrual Cycle, Academic Stress, and Emotional State

1. Motivation:

Why am I working on this project? I guessed that music would be a good mirror of my psyche. As a student, my life is governed by varying degrees of academic pressure, physiological cycles, and sudden emotional shifts. The motivation behind my project was to move beyond verbal evidence (e.g., "I listen to sad music when I'm stressed.") and utilize Data Scien**ce** to quantify these patterns. By integrating my Spotify streaming history with personal logs, I aimed to discover whether my digital consumption could predict my mental and physical well-being, potentially creating a framework for "Mood-Aware" music recommendation systems.

2. Data Source:

Where did I get this data? How did I collect it? My project relies on a multi-source dataset (They cover March 2024 to March 2026, except for Crying Logs—covering January 2026 to March 2026.) integrated at the daily level:

- **Spotify Streaming History:** I have collected via Spotify's "Request Your Data" feature (GDPR). This provided JSON files containing every track played, timestamps, and duration (ms played).
- **Academic Stress Logs:** It is a self-curated CSV file where I have recorded my daily stress levels on a scale of 1 to 5, specifically marking my exam periods as "High Stress" (4 or 5) etc.
- **Physiological & Emotional Logs (converted to CSV files):**
    - **Menstrual Data:** Recorded dates of my cycle to analyze hormonal impact on my mood.
    - **Crying Logs:** Specific timestamps of my emotional incidents.

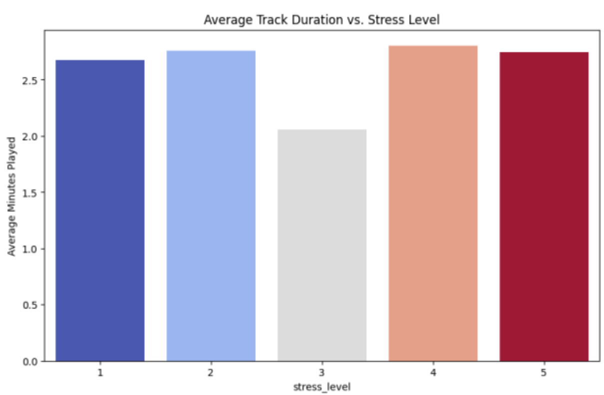
- **Proxy Valence Data:** Since Spotify's API valence was not directly available for all merged tracks, I engineered a "Proxy Valence" feature by mapping genres and artists (e.g., Arabesk, Rock, Pop) to energy scores (0.0 to 1.0) based on musical theory. In order to be more realistic, I have added small randomizations to those scores.

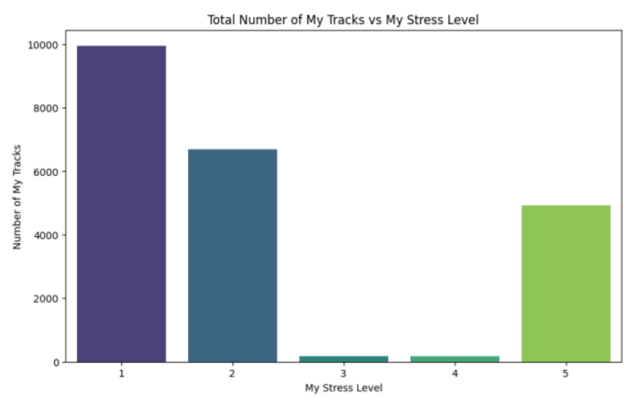
3. Data Analysis

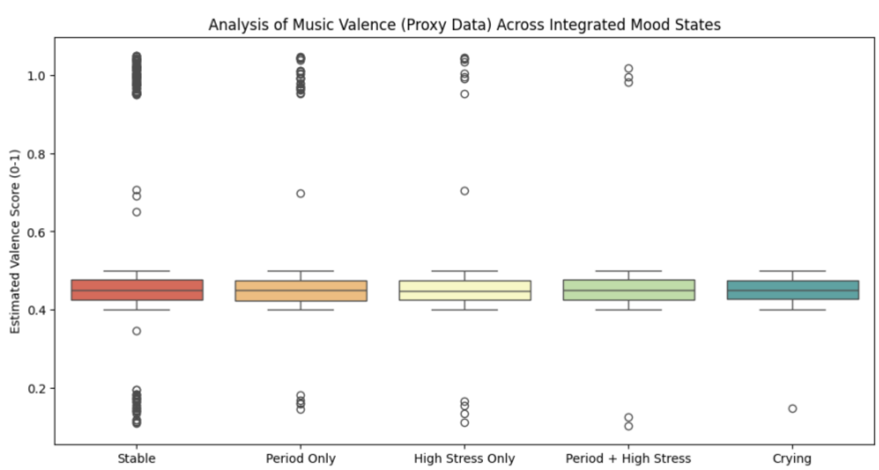
Techniques I have used and different stages of my analysis:

1. **Data Integration:** I have merged multiple JSON sources and synchronized them with CSV logs using a "daily context" mapping function to create a master feature set.

2. **Exploratory Data Analysis (EDA):** I have utilized Seaborn and Matplotlib to visualize the relationships between my stress levels & listening durations, my stress levels & track numbers, and my integrated moods & the valence scores of the songs, as follows:







3. **Hypothesis Testing:**
   - **T-Tests & Pearson Correlation:** These are used to check for quantitative relationships (Duration/# of Tracks vs. Stress).
   - **ANOVA:** It's employed to compare the means of valence scores across my five integrated mood categories (Stable, High Stress, Period, Period+Stress, Crying). (It should be noted that since I have included a randomization factor to determine the valence scores of the songs, the code will produce a different p-value each time it is run.)

These were my 3 hypotheses, respectively:

1- There is a significant relationship between my stress level & average track duration.
2- There is a significant relationship between my stress level & my track number.
3- There is a significant relationship between my (integrated) mood & the valence score of the songs.

4. **Machine Learning:** I have implemented and compared three classification models to predict my "Mood State":
   - **Decision Tree:** For logic visualization.
   - **Random Forest:** For handling variance.
   - **Gradient Boosting (HistGradientBoosting):** Optimized for imbalanced data.

   In this stage, I also made comments on the *visualization of the decision tree* and the *confusion matrices* of all machine learning models I have used.

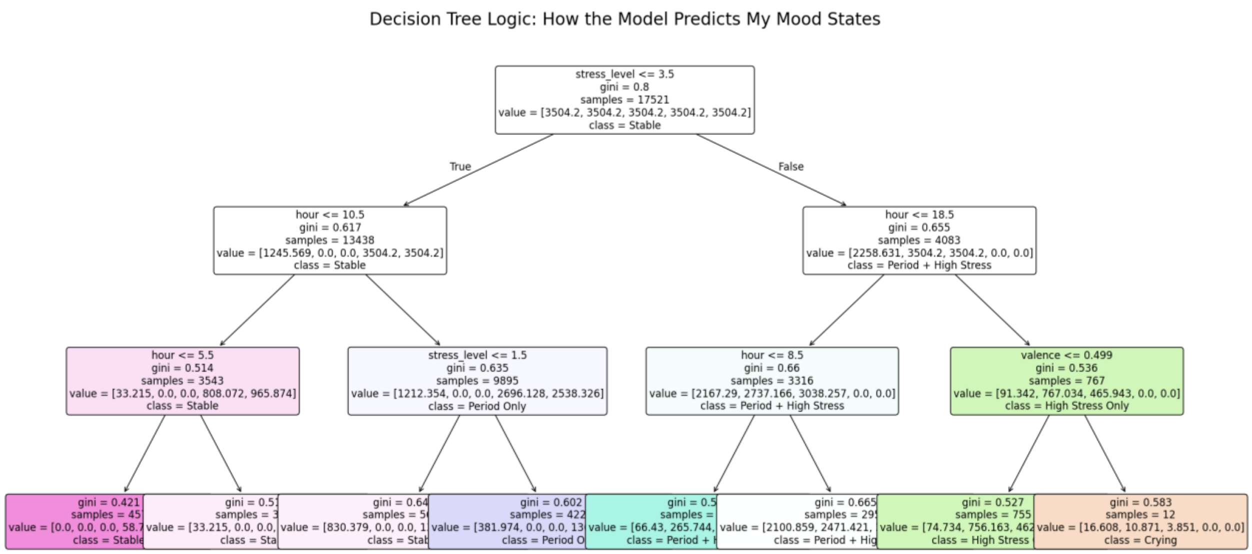
   Furthermore, in order to decide on the best model for my project among them, I have deployed **Weighted F1-Score**. (The winner was "Gradient Boosting" with the highest F1-score.)

4. Findings

What are the key findings of my analysis?

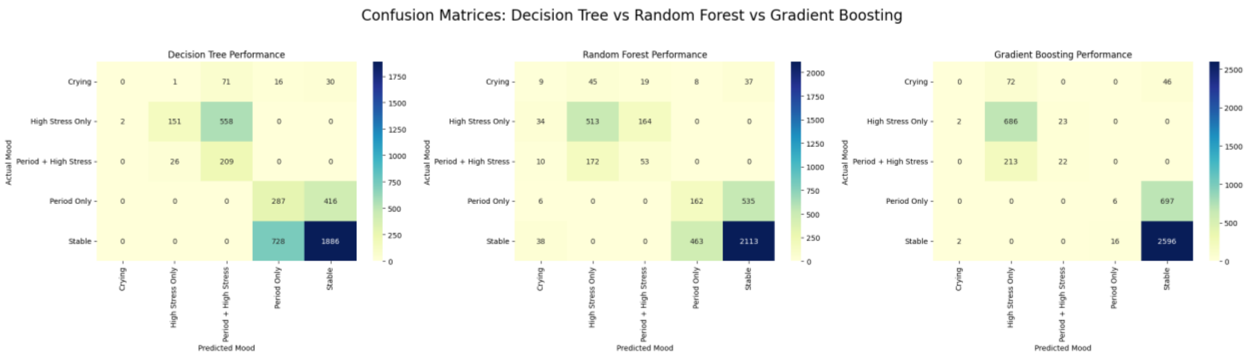
- **The Valence-Mood Connection:** The most significant finding was from the **ANOVA test ($p < 0.05$**, e.g., 0.0142 in the above graph**)**, which proved that my mood state significantly affects the *valence* (emotional quality) of the music I choose, even though the *quantity* (duration/track number) of music did not show a significant change. [The outcomes of the 1st hypothesis test (by t-test) were *T-Statistic: -1.1117, P-Value: 0.2663* **(p>0.05)**. And the outcomes of the 2nd hypothesis test (by Pearson correlation) were *Pearson Correlation: 0.0105, P-value: 0.1208* **(p>0.05).]**

- **Context indicates the Utmost Importance:** The Decision Tree revealed that *stress_level* and *hour* (time of day) are the primary splitters for predicting my mood. This suggests that *when* I listen to music is a stronger predictor of my state than the specific audio features themselves.



- **Model Performance: Gradient Boosting** was the champion model with a **Weighted F1-Score of 0.67** (and **76% Accuracy**).

This is because boosting algorithms iteratively correct the errors of previous trees, which is highly effective for handling the imbalanced nature of mood labels.



Gradient Boosting was particularly effective at distinguishing “Stable” days from “High Stress” days. Unlike the Decision Tree and Random Forest, which heavily confused “Stable” with “Period Only” (728 and 463 misclassifications, respectively), Gradient Boosting minimized these overlaps, demonstrating a more refined decision boundary.

As for the further comparison of the 3 ML models,

Decision Tree: While highly interpretable, it is too simple and makes broad errors, such as mislabeling nearly all “Stable” instances as “Period Only” in certain branches.

Random Forest: Acts as a middle ground. It managed to catch 9 “Crying” incidents (the only model to do so significantly), but its overall accuracy for “Stable” and “Stress” was lower than Gradient Boosting.

Final Verdict: *Gradient Boosting* is the most reliable model for trend prediction, making it the best choice for my project.

- **Temporal Patterns:** The models—as can be traced in the decision tree above—identified a clear distinction in my music habits before and after 10:30 AM, indicating a "routine-based" emotional signature in my listening history.

5. Limitations and Future Work

What could be done better? What future extensions could be explored?

- **Class Imbalance:** A major limitation was the scarcity of "Crying" and "Period" logs compared to my "Stable" days. All models struggled to predict these minority classes.
  - *Future Extension:* Implementing SMOTE (Oversampling), adding more features, or collecting data over a longer period (1-2 years) for minority classes would help balance the dataset.
- **Proxy vs. Real API Data:** I have used Proxy Valence based on artist/genre.
  - *Future Extension:* Using the Spotify Web API (Spotipy) to obtain exact "Danceability", "Energy", and "Acousticness" for every single track would increase model precision.
- **Physiological Sensors:** My stress was self-reported.
  - *Future Extension:* Integrating heart rate data from a wearable (like an Apple Watch) could provide an objective "Stress" feature, removing my probable human logging bias.

6. Conclusion

This project has successfully demonstrated that my digital footprints—specifically my music choices—carry a hidden emotional signature. While predicting rare emotional events like crying remains a challenge for machine learning without more granular data, the ability to identify my high-stress academic periods with over 75% accuracy suggests that data-driven wellness monitoring is a viable and exciting field for future exploration.

7. Academic Integrity & AI Disclosure

I declare that AI tools were utilized during the development of my project. I have limited my AI use to technical assistance, code optimization, and structural guidance for the final report.